

Ruairi O'Hare

 Ruairi O'Hare |  rtboh@icloud.com |  +1 (437)232-7272 | Based in Toronto

EDUCATION

The University of Edinburgh

2017–2021

BSc (Hons) Computer Science, First-Class Honours

- **Dissertation:** Developed a neural network model to predict optimal robot shapes with AI for minimally invasive lung surgery.
- Created a complete SQL interpreter in Java, an autonomous parking robot, and a C-to-MIPS compiler.
- Implemented an LLVM Liveness Analysis feature in C++ and heuristic-based AI games in Java.
- Awarded the **Edinburgh Award** for outstanding on-campus contribution.

TECHNICAL SKILLS

Languages	Java, Python, C, C++, SQL	Backend	Spring Boot, REST APIs, microservices, secure coding, Drupal
DevOps	Docker, Jenkins, CI/CD, Linux, virtual machine deployments, Groovy	Databases	MySQL, Postgres, relational databases, schema design
Testing	JUnit, pytest, mocking, code reviews	Web	JavaScript, HTML/CSS
Frameworks	TensorFlow, Keras, Pandas	Tools	Git, Jira, Confluence, Slack, Overleaf

PROFESSIONAL EXPERIENCE

University of Edinburgh — Lead Digital Preservation Developer

Nov 2022 – Sep 2025

- Built a new digital preservation system from scratch, now used daily by all archivists across the university, significantly automating their workflow and saving hours of manual work daily.
- Developed a pipeline for ingesting large datasets into multiple long-term storage and digital asset management systems; integrating open-source tools, virtual machine management, and long-term infrastructure planning.
- Automated data integrity checks with scheduled runs and email summaries, removing a major manual workload for the archivists and ensuring continuous integrity of archived collections.
- *Tech:* Python, Docker, Linux/VMs, REST/JSON APIs, SQL, Drupal

Adobe — Software Development Engineer

Mar 2022 – Nov 2022

- Built internal code analysis tools to generate quality metrics to evaluate pull requests, applying AI-like insights for project managers to track progress, reduce risk and speed up delivery timelines.
- Integrated these metrics into CI/CD pipelines with Jenkins to automate reporting and enhance deployment speed and reliability using Groovy and Javascript.
- Collaborated with engineers across teams to embed new analysis tools into existing workflows and improve build stability.

Visa Inc. — Java Developer

Sep 2021 – Mar 2022

- Migrated a large legacy platform to a new Java microservice-based architecture for a new Visa digital ticketing product, rewriting and testing core components in Spring Boot to improve maintainability and performance.
- Contributed to Agile development workflows using Docker, Maven, and RESTful APIs, gaining experience with enterprise CI/CD and production deployment practices.

University of Edinburgh — Technical Support and Media Roles

Jun 2020 – Jun 2021

- Supported academic departments with technology procurement, setup, and user training across multiple university projects.
- Edited and published lecture media for online delivery and assisted with large-scale data migration during the transition to remote learning.